

Willian Vieira PhD

Data developer with a PhD in quantitative ecology. I build ETL pipelines, cloud data systems, and reproducible workflows that turn fragmented operational data into trusted, documented, and reusable analytics.

DATA DEVELOPMENT & ANALYTICS EXPERIENCE

2025
|
2024
(1 yr 9 m)

- Data Analyst & developer**
Habitat, Montreal, Canada
Designed cloud/Azure-oriented data infrastructure for organizing heterogeneous project data and making analytical datasets easier to retrieve, validate, and reuse | Architected end-to-end R/Python ETL and analytics pipelines to process large, messy real-world data into decision-ready reports and reusable outputs | Built metadata-driven ETL and provenance workflows to make datasets, definitions, analyses, and handoffs traceable, reproducible, and reliable | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, documentation, and collaborative development practices | Supported stakeholder-facing analytical products with validation checks, documentation, and clear explanations of assumptions and data limits

2024
|
2017

- PhD Research**
Integrative Ecology Lab, Sherbrooke, Canada
Built reproducible pipelines for continental-scale geospatial and environmental data, harmonizing climate, satellite, and field-inventory datasets across North America | Developed Bayesian hierarchical models and simulation workflows to forecast forest dynamics from noisy, biased, incomplete real-world data, with explicit uncertainty propagation and model evaluation | Ran thousands of simulations on HPC infrastructure () to test model behavior, compare assumptions, and scale analyses beyond local compute | Developed open-source software libraries (, ) for demographic modelling with version control, automated testing, documentation, and reusable workflows | Published a **technical methods book** documenting the full computational pipeline, along with one peer-reviewed **publication** and two preprints (, )

2022
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2020
(part-time)

- Biostatistician**
Environment and Climate Change Canada - Quebec, Canada
Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **data pipeline** with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

EDUCATION

2024
|
2017

- PhD, Ecology**
Université de Sherbrooke - Sherbrooke, Canada
 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016
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2015

- Masters 2, Agroecology and Resource Management**
Bordeaux Sciences Agro, Bordeaux, France
 *Modelling the dispersion of weed species in agricultural landscapes*

2015
|
2010

- BSc in Agronomy**
Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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-  w.vieiraw@gmail.com
-  [LinkedIn/WillVieira](#)
-  [GitHub/WillVieira](#)
-  [Publications](#)

SKILLS

Machine Learning & Statistics:

Bayesian inference · Hierarchical models · Uncertainty quantification · Forecasting · Feature engineering

Programming & Database:

Python · R · SQL · Bash · DuckDB · Stan · Julia

Data Engineering & Cloud:

ETL/ELT · Cloud data infrastructure · APIs · Docker · Azure · AWS · GeoParquet · COGs · Apache Arrow · HPC

DevOps & Reproducibility:

Git · CI/CD · Unix · Unit testing · Reproducible pipelines · Automated validation · Provenance

OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught programming, statistics, SQL, and reproducible workflows to 250+ students across biology, engineering, and graduate programs · Translated stakeholder needs into technical analyses, documented pipelines, and clear analytical outputs · Built reusable material that helped non-specialists understand data, assumptions, and results

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional